

Kuan-Hung Chen

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Education

University of Warwick

MSc in Statistics (*with Distinction*)

Focus: *Statistical Inference, Data Analysis, Machine Learning, Neural Networks*

Coventry, UK

Sep 2024 — Sep 2025

National Taiwan University

B.A. in Economics & B.A. in Foreign Languages and Literatures

Double Major, Overall GPA: 3.9 / 4.3. Focus: *Econometrics, Game Theory*

Taipei, Taiwan

Sep 2019 — Jun 2024

Experience

Market Research Intern

KANTAR Worldpanel

Taipei, Taiwan

Jun 2023 — Apr 2024

- Engineered 10+ scalable Python and VBA data pipelines to accelerate the data processing workflow, making Excel formatting and PowerPoint report production at least 20x faster.
- Analysed consumer datasets to extract actionable insights for 10+ FMCG clients, delivering reports and presentations on market trends and demand forecasts.

Research and Projects

MSc Dissertation, University of Warwick

Probabilistic Weather Forecasting with Recursive Algorithms and Vine Copulas

Sep 2025

- Engineered a vine copula-based neural network in PyTorch for mid-term probabilistic weather forecasting.
- Implemented a quasi-Bayesian recursive density estimation algorithm enabling data augmentation without retraining, improving training efficiency.
- The framework proved stable and highly accurate on chaotic time-series benchmarks and real-world WeatherBench data, demonstrating potential for practical application.

Pollparse

ALBERT-based zh-TW poll parser powering Sootow (open source)

2026

- Developed an open-source parser and training framework for Traditional Chinese poll text, tuned to the natural phrasing habits of Taiwanese users.
- Engineered a transformer fine-tuning pipeline on a pretrained ALBERT encoder for span-boundary prediction of poll titles, options, and settings.
- Designed a regex-and-lexicon rule layer converting labelled spans into structured JSON payloads, now serving as the backend API of Sootow, a live polling web app.

zh-TW NLP Product Categorizer

TF-IDF + SVC/MNB classifier served with FastAPI and Docker (open source)

2026

- Built an NLP categorization framework optimised for zh-TW product names that mix English, numerals, and Chinese characters; containerised a production-ready inference API with FastAPI and Docker.
- Achieved 70% accuracy and F1 on highly noisy, error-prone real-world data, establishing a robust performance baseline.

Intaglio

Obsidian-to-Jekyll publishing pipeline, shipped as a GitHub Action (open source)

2026

- Built a Python pipeline that converts Obsidian Markdown into Jekyll-ready output, with incremental builds and asset syncing so only changed notes are rebuilt, and hardened against non-standard real-world Markdown with unit, integration, and CLI-level tests.
- Packaged it as a reusable GitHub Action that publishes to a separate site repository through reviewable pull requests, keeping automated output auditable and reversible.

Skills

Programming: Python, SQL, C++, R, VBA, JavaScript, Ruby

Machine Learning: PyTorch, scikit-learn, NumPy, Pandas, NLP, Transformers, Time Series Forecasting

Engineering & Web: Docker, FastAPI, Git, Bash, GitHub Actions, Ruby on Rails, React, Tableau, Excel

Languages: Mandarin (Native), English (Professional; IELTS Academic 8.0), German (Intermediate)